

A Business Process Simulation Method Supporting Resource Evolution

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ABSTRACT

Business process simulation is the procedure of planning, modeling and simulating enterprise process to analyze the features varying with time which guides the users to make decisions or important means of process improvement. In the existing simulation methods, they considered little about dynamic resource, which may lead to a large deviation. To solve the problem above, a business process simulation method supporting resource evolution that mainly focuses on human resource is proposed. Individual differences and personnel composition are analyzed to represent the dynamic feature of resources. Evolution of human capacity and personnel changes are described to meet the simulation requirements of human resource model. And then, process simulation mechanism and algorithm are realized. An experiment is conducted by simulating the software development process of a real-world software project using the system prototype we developed, and it shows that our method is closer to the practical project situation, thus the method effectiveness can be demonstrated to a certain extent. The main contribution of our work is a novel process simulation approach based on a dynamic resource model with configurable evolution rules.

Categories and Subject Descriptors

I.6.5 [Simulation and Modeling]: Model Development – Modeling methodologies

General Terms

Theory

Keywords

Process model; Resource evolution; Human resource model; Business process simulation

1. INTRODUCTION

Business process simulation is the procedure of planning, modeling and simulating enterprise process to analyze the features varying with time. The existing simulation tools, such as ARIS [1], Simul8 [2], Simcad Pro [3], Protos [4], and EPMS [5], have made researches on process simulation in depth. These tools analyze business process dynamically and provide a series of statistical results to guide users to make decisions or important means of process improvement. The techniques used by these tools fully considered the uncertainty of process model structure, input data and the properties of process objects which leads to an effective simulation result in a certain extent.

As an important part of business process model, the process resource has a strong influence on the quality, efficiency and cost of the process [5]. The resources in actual business process are usually changing dynamically, thus the elements in resource model are evolving constantly, such as the movement of staffs, the growing person ability, the ageing devices and etc. The personnel play a major role in the resources, especially for the software development process which has a high degree of human participation. The personnel information in process model can be obtained from the associated human resource model. The human resource model, which describes the organization structure supporting business process, is a virtual group consisting of persons involved in the process. Nevertheless, in the existing simulation techniques, for example, the Simul8 only takes the number of personnel into consideration [2], and the Protos provides the name of role played and the corresponding available amount [4], and the effectiveness and ability of persons are set as constant in the literature [6,7]. These researches or tools considered constant property and static composition of personnel which reduces the facticity of the simulation procedure and may lead to a large deviation of result.

Addressing the simulation issue of high-degree human involved process, the current techniques of process simulation need to be improved by considering personnel evolution comprehensively to make the simulation closer to the real-world process to obtain a more effective result. Through consulting a relevant literature investigation, a business process simulation method supporting resource evolution that mainly focuses on human resource is proposed. Individual differences and personnel composition are analyzed to represent the dynamic feature of resources. The personnel evolution rules are defined to support the simulation algorithm for the dynamic changes of human resources. Besides, the interaction between resource model and process model is achieved in the simulation executing procedure. To evaluate the

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method we proposed, a prototype simulation tool is implemented. By simulating the real-world software project process with POMES tool, it can be concluded from the experimental data that our method is much closer to the real-world project situation, thus the effectiveness of our method is demonstrated to a certain extent.

The rest of the paper is organized as follows. Section 2 provides the related work of resource evolution of simulation. In section 3 we analyze the overall framework of our method. Section 4 illustrates modeling and simulating of human resources by elaborating dynamic features, evolution rules, and simulation algorithm. In section 5 we discuss the simulation mechanism of process model supporting resource evolution. Section 6 conducts experimental evaluation before Section 7 concludes the article.

2. RELATED WORK

Among the resources in simulation procedure, human resources are considered in this paper. In order to implement the process simulation supporting human resource evolution, the human resource model which is available in simulation should be created first. To satisfy the requirement of dynamic human resource process simulation, we conclude which variable factors should be added into human resource model by investigating the related researches. Currently, most of the researches focus on software development process and typical studies are achieved by Abdel-Hamid [8], Stutzke [10], H. Zhang [12] and Y.X. Chen [13].

Addressing the issue of integrative system dynamics model of software development, Abdel-Hamid [8] constructed the CLD (Casual Loop Diagram) of human resource management in software development process to verify the Brooks' Law which states that adding manpower to a late software project makes it later [9]. The personnel in process are divided into two parts: *new employee* and *experienced employee*. The process transferring a new employee to an experienced one is called *assimilation*. The development process includes the cases of transfer in of new employee and transfer out or resignation of experienced employee. Stutzke [10] divided the new staff into types of *recruiting* and *transferring* in which the new recruited staffs need to be *hired* and *trained* while the transferring staffs only need to be *trained*. Madachy [11] simplified the CLD proposed by Abdel-Hamid to verify the Brooks' Law.

H. Zhang constructed the ASD (Abstract Structure Diagram) of software staffing process to analysis and discuss Brooks' Law using the qualitative simulation model for software engineering process [12]. The personnel are divided into *experienced staff* and *new staff*, and the affection of assimilation process and group communication overhead is considered.

A hierarchy-based hybrid simulation model is proposed by Y.X. Chen who divided software process into two levels: the outer-level model describing the phase changes which used the discrete event model and the inner-level model describing the activity progress which used the system dynamics model [13]. The personnel management is considered as a part of the inner-level model and the improved CLD diagram of developing personnel is proposed.

The variable properties and composing relation of personnel which exist in human resource evolution have been described from different perspectives by the above research. The personnel evolution is abstracted as assimilation process by Abdel and Stutzke only to verify the Brooks' Law and H. Zhang put special emphasis on the effectiveness verification of qualitative

simulation by analyzing the variation trend of personnel. The issue of how to apply the personnel evolution to process model simulation had not yet been clearly considered by these researches. Besides, due to the different research goals, the personnel evolution is usually instantiated to one certain condition which will limit the universality of the methods.

3. METHOD OVERVIEW

In order to achieve the process simulation supporting human resource evolution, we proposed the overall design for our process simulation technique in Figure 1. In the existing method of process simulation, a process model is firstly instantiated to a process model instance which is the input of simulation algorithm. As long as the simulation starts, the states of activities in the process instance are transformed according to the simulation algorithm until the simulation procedure is accomplished and finally obtain a series of data statistic result. During the simulation, the resource model provides kinds of resources, such as personnel, devices, and places while these resources are abstract and static with the only distinguish of being used or not. In our simulation method, the resource model also needs to be instantiated to a resource model instance to simulate with process model instance at the same time, so the state of resources and activities can be changed interactively. In the interaction of simulation, resource model instance provides dynamic resource support to process model instance while the latter triggers the updating of resource in the former through feedback. At this point, we have turned the abstract resource based process simulation into a resource instance based one.

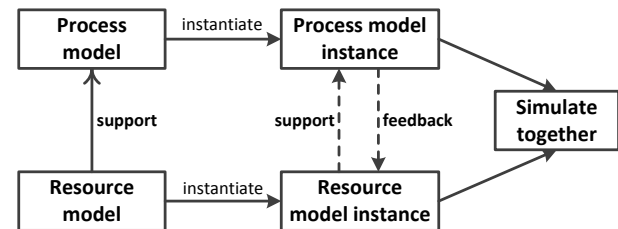


Figure 1. Overall design of business process simulation

In the simulation procedure, to achieve the interaction between process model instance and resource model instance, we need to resolve the problem of how to describe the dynamic variation feature of resource in the modeling phase and also need to present the simulation algorithm for resource model instance. Since we focus on human resources in this paper, we need to discuss the evolution-supported modeling and simulation of human resources.

4. HUMAN RESOURCE MODELING AND SIMULATION

The simulation of human resource model instance includes the resource evolution itself and the interaction with outside environment. In the Figure 2, the human resource model instance runs the simulation algorithm which includes the evolution algorithm and the interaction algorithm with process model. The evolution algorithm runs the resource simulation by resolving the evolution rules and invoking the personnel event simulator. Therefore, we need to firstly analyze the dynamic factors of personnel in resource modeling and then obtain the simulation algorithm through the evolution rules which describe these factors to achieve the human resource simulation finally.

4.1 Human Resource Modeling

The existing modeling tools, such as EPMS [5], include the model elements of organization structure and roles. The movement of personnel doesn't exist in organization structure and there are only the differences of name and number between the various roles. When we consider the dynamic features of personnel, from the individual viewpoint, personnel differences in resource model emerge due to the ability changes of personnel. On the other hand, from the overall viewpoint, adding newbies or staff movement will affect the quantity and composition of personnel in resource model. Therefore, individuals should be distinguished in roles and personnel movement should be described in organization.

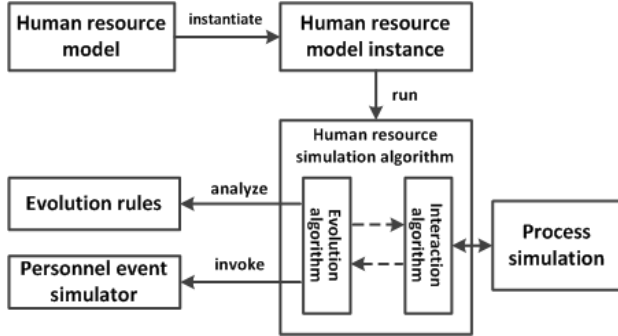


Figure 2. Human resource simulation

4.1.1 Individual Differences

In process simulation, individual differences of personnel property influence the activity handling and the person who does not reach certain ability will not participate in activities. Therefore, to distinguish personnel type for a certain business process, the personnel are divided into *experienced staff* and *new staff* according to their ability level. The new staff includes two categories of *new recruited staff* (new staff from outside the organization) and *transfer staff* (new staff transferred from other department in the organization). The procedure turning new staffs to experienced ones is called *training process*. The ability of personnel is characterized by *ability value* and its variation fulfils the requirement of individual evolution.

The *ability value* is the workload of a person acting as a certain role in per unit time. Note that the measurement criteria of workload vary with different types and stages of process, e.g. in software development, the workload of requirement analysis can be measured by document pages and the coding phase can be judged by LOC. The *workload* includes three parts: activity handling, training, and collaborative communication. The *effective ability value* is the activity handling workload of a person acting as a certain role in per unit time. The initial ability value of experienced staff is denoted as V_{min} and the upper bound is V_{max} . The initial ability value of new recruited staff and transfer staff is 0 and V_{trans} respectively (V_{trans} is a value between 0 and V_{min}).

In the simulation procedure, a staff can only handle one activity at one time, thus availability property is added to personnel to distinguish whether personnel are being used. Finally, the personnel properties describing individual differences can be concluded as: type, ability value, and availability.

4.1.2 Personnel Composition

From the viewpoint of ability value, the personnel in human resource model are composed of experienced staff and new staff,

and the increase or decrease of them will influence the composition of personnel. As shown in Table 1, the recruitment will increase the number of new staff while the resignation would reduce the number of experienced staff. Experienced staff from different departments can be transferred which will affect the ratio of different staff types. The training process will reduce the number of new staff and meanwhile increase the number of experienced staff. Therefore, we obtained three types of personnel movement: recruitment, transfer, and resignation.

When the recruitment of personnel happens, the training process will be triggered at the same time, thus the training strategy should be taken into consideration, i.e. how to select experienced staffs to execute training, how does the activity handling ability (i.e. effective ability) change to these training-involved staffs, what is the variation law of the new staff's ability, and how does the ability recovery after the training end etc. For the personnel transferring, due to the different working contents, the training process is also necessary for the transfer staffs which will be different from the new recruited ones' training. All these above aspects of issues will be illustrated by our simulation method. Note that for these personnel movement events, parts of them are predictable (e.g. regular recruitment from the society or the graduation of school students) while some others are only probably happened in a specific time period, thus both of the definitive and random personnel movement events should be supported by our simulation.

Table 1. Personnel composition

Variation	New staff	Experienced staff
Increase	Staff recruitment	Turn to experienced staff after training
	Staff transferred in	
Decrease	Turn to experienced staff after training	Staff transferred out
		Staff resignation

4.2 Human Resource Simulation

Based on the description of personnel dynamic factors in human resource modeling above, the corresponding personnel evolution rules can be obtained. The personnel evolution algorithm simulating dynamic variation of resources can be achieved by resolving these rules. Besides, the interaction algorithm, from the external view, can support the personnel information interaction with process model simulation; on the other hand, personnel information synchronization with internal evolution algorithm is implemented.

4.2.1 Configurable Personnel Evolution Rules

Based on the dynamic evolution factors of personnel achieved in section 4.1, we can define the personnel evolution rules which describe how the evolution happens in detail.

When a new staff entered the project group, a training process is required to study the background, processing flow and techniques of the project before taking part in activity handling as an experienced staff. Due to the different original ability and experience of various people, the training duration is different, but generally speaking, new staffs will be more experienced with the advance of time. New staffs will be trained by project-related content and they will not directly involve in the project activity

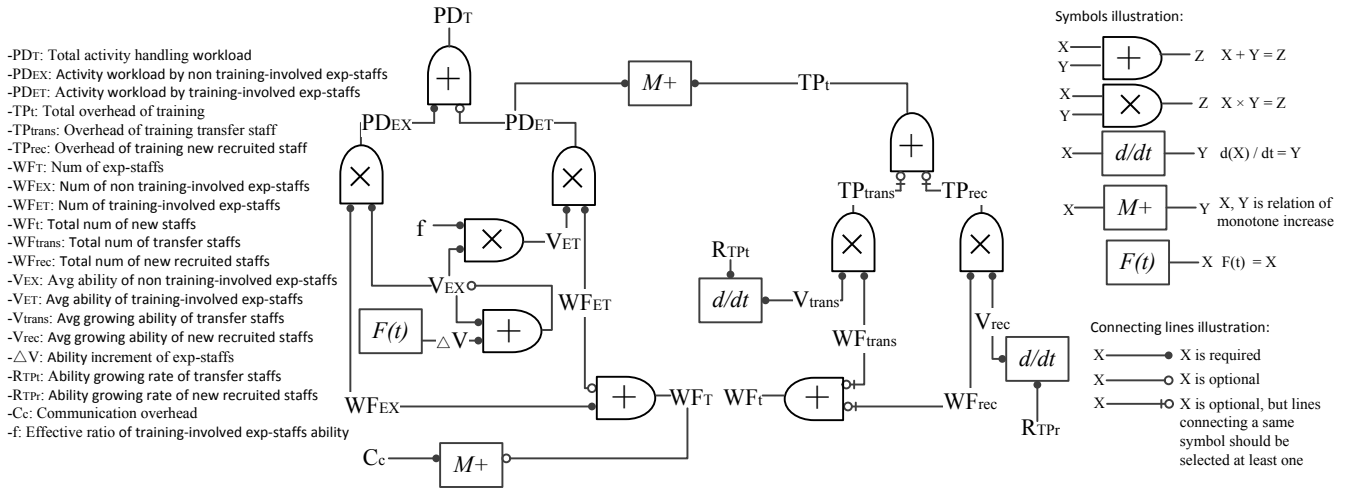


Figure 3. The configurable computing relations of personnel evolution

Table 2. Calculating expressions of personnel evolution

ID	Elements	Graphical representation	Description
1	Number of experienced staffs not involved in training process (WF _{EX}); Number of experienced staffs involved in training process (WF _{ET});		Non training-involved exp-staffs must be included, if training process exists, then the training-involved exp-staffs should be included.
2	Number of new recruited staffs (WF _{rec}); Number of transfer staffs (WF _{trans});		Types of new staffs: new recruited staffs and transfer staffs. At least one type should exist.
	Overhead of training new recruited staff (TP _{rec}); Overhead of training transfer staff (TP _{trans});		Total overhead of training-involved exp-staffs may include two parts.
3	Total Activity handling workload (PD _T); Activity handling workload by experienced staffs not involved in training (PD _{EX}); Activity handling workload by experienced staffs involved in training (PD _{ET});		Activity handling workload by exp-staffs not involved in training must be included, if training process exists, then workload by exp-staffs involved in training should be included.
4	Activity handling workload by experienced staffs involved in training (PD _{ET}); Total overhead of training (TP _T);		The more activity handling workload by exp-staffs, the less overhead is spent in training.
5	Average ability of experienced staffs not involved in training (V _{EX}); Average ability of experienced staffs involved in training (V _{ET}); Effective ratio of exp-staffs involved in training (f);		Effective ability decreases due to the training overhead, and the decreasing ratio can be configured.
6	Ability increment of experienced staffs (ΔV);		The ability value of exp-staffs activity-driven increases according to a specified function which can be configured.
7	Average ability grow of new recruited staffs (V _{rec}); Ability growing rate of new recruited staffs (R _{TP_{rec}}); Average ability grow of transfer staffs (V _{trans}); Ability growing rate of transfer staffs (R _{TP_T});		Ability value of new staff grows linearly with time in training process.
8	Communication overhead (C _c)		Communication overhead is related to quantity of participants of an activity.

handling to ensure the quality of project. According to the above analysis, the follow rule is achieved:

Rule 1. The training process starts immediately when new staffs (i.e. new recruited staffs or transfer staffs) join in. In the training process: 1) The ability value of new staff is time-driven growing; 2) The ability value of new recruited staff is growing slower than the transfer staff due to the transfer staff is more familiar with culture and operational process of the enterprise, which will result in a shorter training period; 3) The experienced staff with highest ability value is preferentially selected to train new staff; 4) The ability value of new staff reaches the initial value of experienced staff after training process; 5) The new staffs do not handle activities in the training period.

With the advance of time involved in project, the more content of activities handled by the experienced staffs, the more familiar skill they will get (e.g. the higher efficiency of coding) until their ability reaches a stable value. Therefore, we can obtain the second rule as follows:

Rule 2. The ability value of experienced staff is activity-driven growing. When an experienced staff accomplishes an activity, his ability value will increase according to a specific functional relation until it reaches the upper bound V_{max} .

The training process results in the ability value changing of both experienced staffs and new staffs which means the new staffs improves their ability through training while the efficiency of activity handling by experienced staffs decreases due to the strength scatter in training new staffs. The above situation can be concluded as follows:

Rule 3. The training process has an effect on the effective ability value of experienced staff. The effort of experienced staff is partly used to train the new staff which will lead to a decrease of effective ability value. Since the more efforts spent to train the new recruited staffs than transfer staffs, when training different types of new staffs, the decreasing ratio of effective ability value is also different. The literature [12] mentioned that the training of a new staff needs to take 15%-25% workload of an experienced staff while Madachy [11] consider this value to be 25%.

Besides, the group communication is required to be considered in the simulation. Although adding more persons to an activity will share the workload, the interaction between them will be more frequent and complicated. Thus, the last rule can be concluded.

Rule 4. Interactive communication in group affects the effective ability value of experienced staff. The group communication overhead exists in the experienced staffs who participate the same activity, so the spending value is related to the number of participants (N). The literature [9] claimed that this value is proportionate to N^2 , i.e. the more participants in one activity, the greater decreasing of effective ability value of them.

We use the ASD (Abstract Structure Diagram) [14] which is frequently-used in process simulation to clearly describe and visualize the computing relationship of personnel evolution and the interaction between the evolution rules. The relation of *plus* ($\oplus$) and *multiply* ($\otimes$) can be easily described in ASD, and other relation such as minus or divide can be derived from them. The uncertain dependency relation can be expressed by *monotone increase* ($\overset{+}{\square}$) or *monotone decrease* ($\overset{-}{\square}$). According to the different requirements of real-world business process, we extend the ASD by adding three composition relations: *required* ($\rightarrow$), *optional* ($\rightarrow^o$) and *at least one selected* ($\rightarrow^{\circ}$); and one *function*

relation ($\overset{F(t)}{\square}$). The configurable computing relations of personnel evolution is shown in Figure 3. The value of elements in the figure is changing continually over time according to the evolution rules. The Figure 3 is decomposed into calculating expressions in Table 2.

Table 3. Evolution rules supported by researches

Researches	Rule 1		Rule 2	Rule 3	Rule 4
	Add staff recruited	Add staff transfer			
Abdel	✓			✓	✓
Stutzke	✓	✓		✓	
H. Zhang	✓			✓	✓
Y.X. Chen	✓			✓	

The evolution rules should be customizable for users to meet the different requirement of various business processes. The customization concept of rules can be divided into two levels: configurable rules or extendable rules. The former means that the users select and configure from a pre-set collection of rules while the latter allows users to expand new rules to the existing collection. By analyzing the adjustable portion of evolution rules firstly and then defining configurable points for these rules, the concept of configurable rules is adopted in this paper which guarantees the correctness and a certain degree of flexibility.

Table 4. Configurable items and expressions of rules

Rule	Config item	Expression	Description
1	New staff composition	$WF_t = aWF_{trans} + bWF_{new}$	$a = 0$ or $a = 1$ $b = 0$ or $b = 1$
	Ability of new staff	$R_{TP} = d(V_{trans})/dt$	Time-driven growing
2	Ability of exp-staff	$V_{EX} = V_{EX} + \alpha \Delta V$	$\alpha = 0$ or $\alpha = 1$
		$\Delta V = F(R_{exp})$	Activity-driven growing
3	Ability of exp-staff involved in training	$V_{ET} = V_{EX} * f$	$f \in (0, 1)$
4	Communication factor	$C_c = \beta(WF_T)^2$	$\beta \in (0, 1)$

In the personnel evolution rules, the Rule 1, Rule 2, Rule 4 is optional, the Rule 3 is required when the Rule 1 is selected, i.e. we can set the following simulation cases: any new staff join in or not, increasing of experienced staffs' ability or not, the ability of experienced staff involved in training process decreased or not, and group communication overhead exists or not. Abdel-Hamid, Y.X. Chen and Stutzke used the Rule 1 and Rule 3 in which the former two considered the adding of new recruited staffs while the latter also considered the transfer staffs. H. Zhang used the Rule 1, Rule 3 and Rule 4. Thus, Table 3 can be obtained by the above analysis and it can be concluded that the personnel evolution rules proposed by this paper can support the dynamic description of personnel in the existing researches. Each evolution rule corresponds to different configurable items which can be described by expressions in Table 4. After selecting rules and

setting related coefficient of expressions, the personnel evolution rules available in simulation can be finally achieved.

4.2.2 Simulation of Human Resource Model

The simulation algorithm of human resource model includes two parts: the evolution algorithm simulating the personnel evolution and the interaction algorithm with process model. These two parts communicate with each other in time to ensure the synchronization of personnel status information. The operation procedure of the algorithm is illustrated in Figure 4.

The evolution algorithm analyzes the personnel evolution rules which have been configured and handles the personnel movement events generated by personnel event simulator to guarantee the dynamic variation of personnel of human resource model instance. List of personnel information and personnel events are inputted into the algorithm. The personnel event simulator can generate the personnel events of recruiting, transfer and resignation randomly or by the pre-set values according to empirical data. When the simulation begins, the evolution algorithm starts the ability maturing process of experienced staffs according to Rule 2 and handles the affection of group communication based on Rule 4. Advancing by the simulation clock, the evolution algorithm detects if there is any event to be handled, and if there is new staff of recruiting or transfer, the training process of Rule 1 is activated and the effective ability value of training-involved experienced staff is computed by Rule 3. When resignation happens to experienced staffs, the related information in personnel list of the corresponding staff will be deleted. The evolution algorithm handles personnel movement events continually until the end of process simulation.

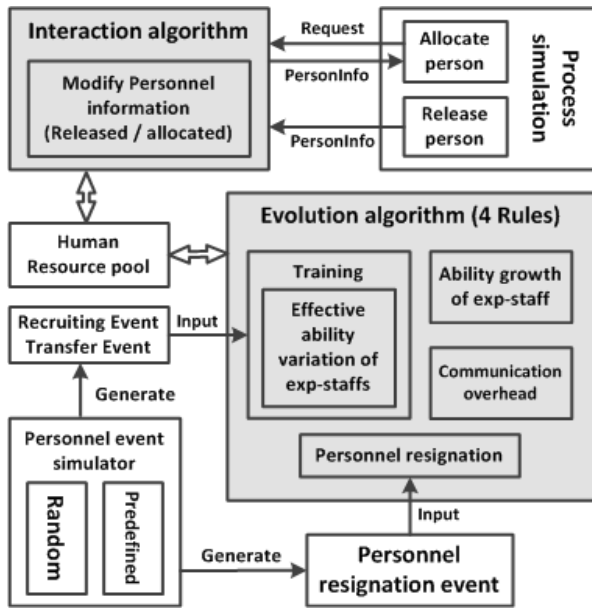


Figure 4. Simulation algorithm of human resource model

The interaction algorithm uses the personnel property of type, ability value, and availability as input data. In the simulation process, when there is a request from process model, the algorithm accepts the request and handles as following: 1) If it is a personnel request, the algorithm calculates the current personnel information according to the time interval and previous personnel status of the last request and allocates persons for process model with the unavailable-marking of these allocated persons; 2) If it is a personnel release request, the algorithm computes the ability

growth for the released persons according to the feedback information by process model and marks these persons as available.

In the simulation procedure, when the personnel movement event triggers evolution algorithm, the evolution algorithm provides updated personnel information for interaction algorithm; and when request from process model triggers interaction algorithm, the interaction algorithm notifies the evolution algorithm to adjust the related personnel information. Therefore, the personnel information synchronization of evolution algorithm, interaction algorithm and process model invoking is ensured.

5. PROCESS SIMULATION SUPPORTING HUMAN RESOURCE EVOLUTION

The process model and human resource model should be constructed and instantiated before the simulation execution which includes the process modeling, modeling of organizations and roles, configuring evolution rules, setting of relevant parameters and etc. The process simulation architecture supporting human resource evolution is illustrated in Figure 5. The process simulator, controlled by simulation engine, executes process model instance by model interpreter and carries out the event handling, resource scheduling and data analysis. The random number generator provides random data for simulation and simulation clock offers timing control and time step. The resource evolution supported part is reflected in the invoking of human resource simulation algorithm when the resource scheduling. The synchronization of personnel information is ensured by simulation clock invoked by the human resource simulation algorithm in which the evolution algorithm analyzes the evolution rules and calls the personnel event simulator to imitate the dynamic variation of persons while the interaction algorithm handles the personnel information when the request or release event of resources happens.

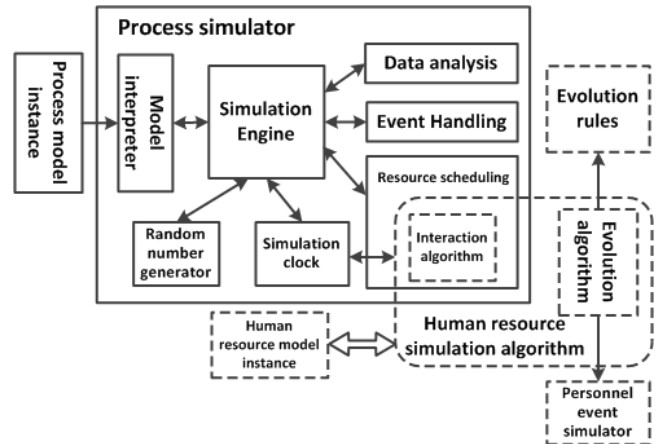


Figure 5. Architecture of process simulation

5.1 Executing Strategy of Process Simulation

Based on the event-trigger mechanism and queuing model, the execution of process simulation constitutes a discrete event system. The executing process is based on the event which is generated by the entities in simulation system, i.e. products and resources in business process. The events generated by the state changing of products include:

- 1) Product Generating Event (PGE) is produced when a product generates (i.e. arrival of source product or generating of

output product) and the event will be put to all of the corresponding output dataflows.

- 2) Activity Ready Event (ARE) is produced when all input products are available, i.e. each input dataflow of the activity contains at least one PGE.

The events generated by the state changing of resources include:

- 1) Resource Releasing Event (RRE) is produced when activity is completed and the state of the resource should be updated.
- 2) Resource Using Event (RUE) is produced when all the resources required by the ready activity are satisfied and allocated which means the activity execution is started.
- 3) Activity Executing Event (AEE) is produced when activity handling starts, i.e. resources of the activity are allocated.
- 4) Activity Finished Event (AFE) is produced when the activity achieves the specific workload and the resources occupied will be released immediately, i.e. RRE is produced.

For the discrete event simulation, the basic elements are events, activities and processes which are closely related to the variation of system state. The frequently-used execution strategies include Event Scheduling (ES), Activity Scanning (AS) and Process Interaction (PS). There are probably multiple ready activities at the same time in the simulation and the activity handling duration is not constant, thus the PS is adopted which integrates the benefit of ES and AS. The Current Event List (CEL) and Future Event List (FEL) are used in simulation. In the simulation procedure, the CEL is scanned continually to handle all the activities satisfying the resource condition and then put the events generated by these activities into the FEL. After that, the simulation clock will be advanced forward to next nearest time point in FEL and the event of this time will be moved into the CEL.

The simulation mechanism is described by the generating, queuing and consuming of events which are activated at the dataflows. If a new event arrives before the old events have been consumed, the new event should queue at the dataflow. Each time of activity execution will reduce the queue length of each input dataflow of this activity by one. As shown in Figure 6, the activities in process model continually change the states of initial, ready, executing and finished triggered by kinds of events, meanwhile, produce and consume these events.

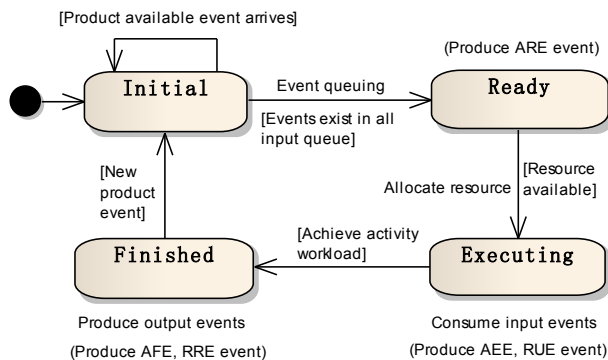


Figure 6. State transition of the activities

All the activities in process model instance are in the initial state when the simulation starts. When an event generated (e.g. product arrives), it will be put to the queue of the corresponding connection. The activity will turn to a ready state when event exists in all of its input connections and then send request for

personnel to human resource model. The process model instance then selects persons according to the allocating strategy based on the personnel information list provided by the resource model instance. If persons are allocated to the activity successfully, the information (i.e. availability) of the corresponding persons will be updated in resource model and the activity execution starts. In the meanwhile, triggered by the personnel evolution events, the evolution algorithm will update the personnel information continually.

5.2 Scheduling and Allocating of Resources

From the simulation executing strategy mentioned above, a specific scheduling policy is required to deal with the ready activity pool to arrange the access to the resources which determines the executing order of these ready activities. Besides, how to allocate the appropriate personnel for a certain activity is also need to be considered.

In process simulation, only experienced staffs can be allocated to handle activities and each person owns a property structure of role, type, availability and ability value. The activity parameters include total workload, duration restriction, number of persons required restriction and etc. Total workload means the workload summation of all roles required. The duration restriction can define the upper bound of time of the activity. Besides, there may be a minimum requirement of persons in a certain role involved in the activity. The personnel allocating policy can be concluded as follows: 1) Check the property of type and availability in the personnel list to obtain the alternative person set P ; 2) According to the restriction of minimum persons of the roles, select alternative person groups P_{ROLE} for each role; 3) Select persons from each P_{ROLE} which satisfy the maximum duration restriction under the premise of reaching the total workload of the activity.

Algorithm 1. Requesting and releasing of personnel

INPUTS: current time T , last request or release time t , person info s , request queue $reqQueue$

OUTPUTS: personnel information s

```

1  interval = T - t
2  s1 = calculate(s, interval)
3  update(s, s1)
4  // People Allocate
5  IF reqQueue.type INSTANCEOF allocate
6    newReqQueue = sort(reqQueue)
7    FOR i = 1 to reqQueue.size
8      IF isAvaliable(newReqQueue.get(i))
9        s2 = allocate(newReqQueue.get(i))
10     update(s, s2)
11  // People Release
12  ELSE IF reqQueue.type INSTANCEOF release
13    FOR j = 1 to reqQueue.size
14      s3 = release(reqQueue.get(i))
15      s4 = calculate(s3, reqQueue.get(i).interval)
16      update(s, s4)
17  t = T

```

Various scheduling rules have been proposed according to the different types of business process and a combining policy of HPFS (Highest Priority First Service), FCFS (First Come first Service) and RS (Random Selecting) are used in this paper. If the same person is request from more than one ready activity, the person is allocated according to the following scheduling strategy: 1) First serve to the activity with highest priority; 2) If multiple

activities have the same highest priority, serve to activity with the longest waiting time; 3) Random selecting in the activities with the same priority and waiting time. When activity executes, the number of events in each input connection will be reduced by one. When activity finishes, the personnel executing this activity will be released and each output connection will generate one event to activate the succeeding activities. In the meanwhile, the resource model instance adjusts the information (i.e. ability value and availability) of released persons. The Algorithm 1 illustrates the process of personnel requesting and releasing.

In the Algorithm 1, the information of new staff in training process is firstly updated (Line 1~3). When personnel request happens, the requests are sorted according to the scheduling strategy (Line 6). The requests are checked one by one according to the queue sequence, and if the request is satisfied, personnel are allocated for this request (Line 7~9). The resource model instance will be triggered to update personnel information after each allocation (Line 10). When personnel release, the ability value and availability of released persons are updated by the resource model instance (Line 13~16). When these operations are done, the last request or release time t is set as the current time T (Line 17).

6. EXPERIMENTAL EVALUATION

To evaluate the effectiveness of our method, we simulate a real-world project which is the software development process of Ships Comprehensive Condition Assessment and Health Management System undertaken by our research group.

The process model and human resource model of this project were constructed by the prototype tool POMES (Enterprise Process Modeling and Optimization System) developed by our research group which is shown in Figure 7. The process model included 5 stages of requirement analysis, system design, coding, system testing, and delivery which amount to 69 activities with 16 milestones in them. The human resource model contained 5 types of roles including project leader, analyst, designer, developing engineer and testing engineer and the project leader was one PH.D candidate and 6 master students served as the other roles. After setting simulation parameters according to the project workload, personnel quantity and project plan, the average simulation result of project total period is achieved as 411.3 days by the simulation tool without resource evolution.

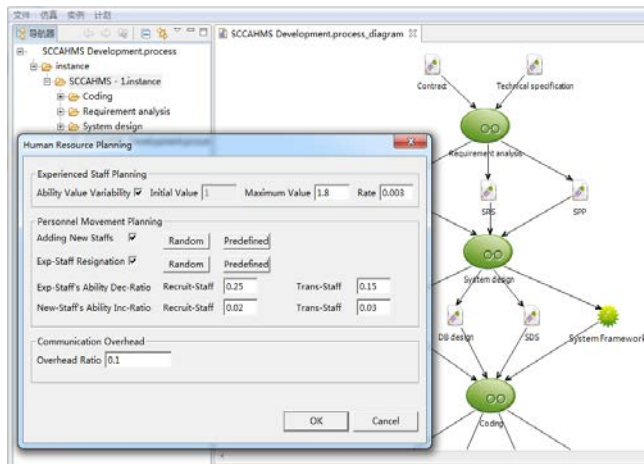


Figure 7. Process simulation prototype POMES

The personnel evolution happened in the actual implementation of project. Two new master students added to the role of developing

engineer at the term beginning while one master student quitted because of graduation at the end of the term. Besides, due to the high workload in the coding and system testing phase, two master students transferred into this project group from other groups.

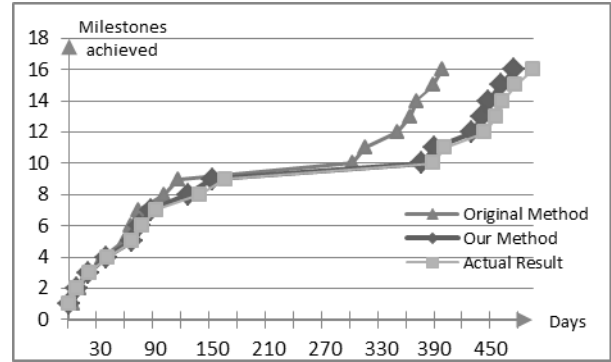


Figure 8. Project progress comparison of simulation results

This software development process accorded with the process simulation supporting resource evolution. The human resource model should first be instantiate according to our method. According to Table 4, the configuration (details can be achieved in Figure 7) can be set as: personnel included experienced and new staff; new staff contained new recruited and transfer staff; the ability value of experienced staff grew linearly with activity handling time; the ability decreasing factor of experienced staff involved in training is set as 25%; the group communication overhead is considered. Besides, the foreseeable movement (i.e. adding of new students and graduation) of personnel and random events are set to the personnel event simulator. The simulation run after all these parameters set properly and the result of project total period is 483.6 days while the actual duration of the project execution is 497 days.

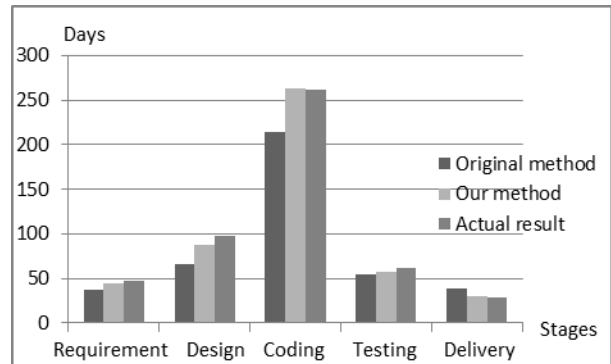


Figure 9. Total duration of each stage

The project progress comparison of the simulation results of **original method**, i.e. simulation result without resource evolution, and **our method** which considered the personnel evolution in software development process, and the **actual result** of project execution is shown in Figure 8. The abscissa represents passage of time while the ordinate represents the accomplishment of project which is measured by the number of milestones achieved. From the figure we can see that: 1) The lines are not significantly different at the beginning, but divergence between original method and our method appears from the 4th milestone; 2) The time of activity handling is extended from milestone 6~8 in our method due to the graduation of a master student; 3) Our method considered two transferred master students during the milestone 8~10 and the effective ability value of experienced staffs will be

decreased due to the training in this period. 4) The training process has finished during the milestone 10~12, so the number of experienced staffs increased and the time of activity handling decreased.

Besides, we can see that for each milestone, our simulation method is more coincident with the actual execution. The total duration of each project stage is shown in Figure 9. From this viewpoint, we can see that the significant difference between the two methods is shown in the coding phase, and our method is still more coincident with the actual execution from the stage-duration perspective.

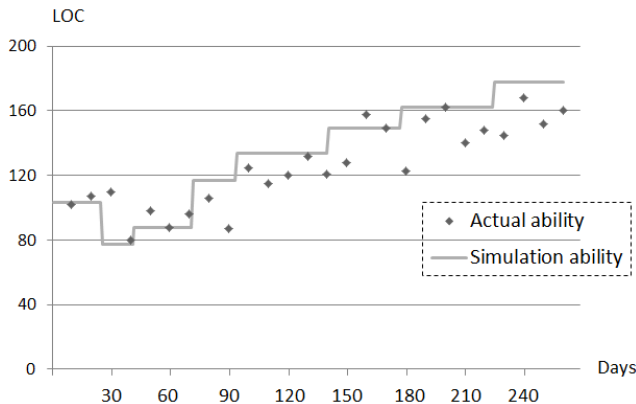


Figure 10. Average ability variation of experienced staffs

There are two training processes in this simulation procedure: the attending of new students and the transferred students from other groups at the coding stage. Here we consider the variation of average ability value of training involved experienced staffs which is shown in Figure 10. The abscissa represents passage of time while the ordinate represents the ability value which is measured by the lines of code everyday recorded by the software configuration management tool SVN. Note that the average values of actual LOC in every 10 days are pointed in this figure to avoid too much unstable data. From the figure we can see that: 1) The ability variation of the staff largely conformed to the activity-driven increasing rule proposed by our method; 2) The effective ability value decreased when the staff involved in training during the period about 30th~80th days. 3) The ability value will not change significantly when it reaches the upper bound (i.e. from about 220th days later).

7. CONCLUSION

To resolve the deviation of simulation result caused by the lack of consideration of dynamic resources in the existing process simulation methods, a business process simulation method supporting resource evolution which is focused on the personnel variation was proposed in this paper to make the simulation result more effective. The individual differences and the dynamic variation of personnel composition were first considered based on the existing research survey and then the configurable personnel evolution rules of human resource model simulation were constructed. Then the mechanism, algorithm and architecture of process simulation were presented. Finally, by an experimental simulation of a real-world software development process based on the prototype implemented, it has been shown that our method is

more coincident with the actual process execution which demonstrated the effectiveness of our approach to a large extent. Therefore, the main contribution of our work is a novel process simulation approach based on a dynamic resource model with configurable evolution rules.

In the future research, we aim to extend the evolution rules to describe more variation factors in process simulation. Our method will be applied to more industry software projects to further demonstrate the effectiveness of our approach. How to measure the personnel ability value more precisely in different phase of process is another issue to be considered. In addition, evolution of other resources except for personnel can be taken into consideration referring the approach of this paper in the future.

8. ACKNOWLEDGMENTS

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